HAMED AFZALI

Senior Full-Stack Engineer | Technical Lead | Software Architecture & Distributed Systems

Tübingen, Germany | +49 176 3146 1176 | afzali.hamed@gmail.com | linkedin.com/in/hamed-afzali | afzali.eu.cc

SUMMARY

Senior full-stack engineer and technical lead with 20+ years of experience designing, modernizing, and running software platforms in regulated banking, fintech, and SaaS environments. Led an 8-person engineering team for 9 years and was the sole architect of a regulatory analytics platform used across 400+ branches and for Central Bank reporting.

At GMG, works across architecture and delivery on a multi-tenant SaaS platform: leads the .NET 8 migration, designed the security and identity layer (OAuth2, JWT, fine-grained authorization), helped shape the CI/CD setup, and built AI-assisted engineering tooling (MCP, RAG) and CI-enforced licence compliance.

Makes and documents technical decisions, sets engineering standards, works closely with stakeholders, and still writes production code in C#/.NET, TypeScript/React, Node.js, and SQL Server.

SKILLS

Architecture & Design: Distributed Systems, Microservices, Event-Driven Architecture, Domain-Driven Design (DDD), API Design (REST, OAuth2, JWT), Architecture Decision Records, Legacy Modernization, UML, Business Process Modelling

Leadership: Team Leadership (8 engineers, 9 years), Architecture and Code Reviews, Engineering Standards, Stakeholder Alignment in SAFe Agile, Mentoring, SOLID, OOA/OOD, TDD, GDPR-aware development

Languages & Frameworks: C#, .NET 8, .NET Framework, TypeScript, JavaScript, React, Redux, Node.js, Python, Java, C++

Data & Messaging: SQL Server, MongoDB, SQLite, NHibernate, ETL, IBM MQ

DevOps & Tools: Docker, CI/CD, Azure DevOps, AWS (CloudWatch), Cypress, Jira, Git

AI Engineering: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), AI Agents, Model Context Protocol (MCP), LangChain, LangGraph, Vector Databases, AI-assisted Software Development

PROFESSIONAL EXPERIENCE

Senior Full-Stack Engineer

GMG GmbH & Co. KG | Tübingen, Germany | 2022 - Present

Multi-tenant SaaS platform for professional print hardware, in a SAFe Agile team. Work covers architecture, backend, frontend, infrastructure, and hardware integration.

- Designed and implemented the security and identity layer: REST APIs with OAuth2 and JWT authentication, and fine-grained authorization with Authzed for multi-tenant permissions under GDPR data-protection requirements.
- Helped make the decision to move from .NET Framework to .NET 8 and leads the ongoing migration, including moving local storage from SQL CE to SQLite, while the product stays stable in production.
- Helped define the CI/CD approach and set up pipelines in Azure DevOps with Docker, covering build automation, release workflows, and containerized deployment.
- Designed, built, and maintains a Python MCP server with a RAG index of the codebase (since 2024), so LLM assistants can search and explain the code. This has made onboarding to the platform much faster.
- Independently designed and built (2023) an automated OSS licence-compliance tool for 3 applications and their transitive dependencies (57+ NuGet packages in the main app). It checks each package against the licence policy, generates copyright notices, an acknowledgement file, and an SBOM, and blocks unreviewed packages in CI. Manual legal review is no longer needed.
- Designed and built the IPC layer between the software and the printer hardware using Named Pipes and sockets, giving reliable real-time communication and removing race conditions.
- Built and maintained automated tests (unit, integration, Cypress E2E), which reduced production bugs, and helped set up early AWS CloudWatch monitoring.
- Delivers React and TypeScript features for the multi-tenant UI and works with stakeholders to balance new features against technical-debt reduction.

Technical Lead, Analytics Platform

Postbank of Iran (PBI) | Tehran, Iran | 2018 - 2022

Sole architect and technical lead of a regulatory reporting and analytics platform that consolidated data from 400+ branches for senior management and Central Bank submissions.

- Designed and delivered a BI and reporting platform (React.js, Node.js, Docker, microservices) that gave management and Central Bank regulators one real-time view of core banking, 400+ branches, ATMs, and POS, replacing manual reporting.
- Owned the platform end to end as a self-hosted, on-premises deployment, including infrastructure, configuration, and production operations, and was the 2nd-level technical escalation for 400+ branches under banking SLAs.
- Designed Node.js and SQL Server ETL pipelines with strict validation rules, because errors in regulatory data had legal and financial consequences for the bank.
- Added an API layer for external BI tools and designed a React Native app for branch-level reporting.

Technical Lead, Core Banking R&D

Postbank of Iran (PBI) | Tehran, Iran | 2009 - 2018

Grew from senior developer to technical lead of an 8-person engineering team building core banking systems and national payment infrastructure for one of the region's largest financial institutions.

- Built and led a cross-functional team of 8 engineers for 9 years, responsible for delivery, sprint planning, code quality, and mentoring.
- Designed a full core banking platform (.NET, C#) covering Accounting, Treasury, HR, Branch Operations, and Supervisor workflows. Modeled the processes with the banking departments in UML before implementation. The platform ran on a 15-node SQL Server cluster serving 15,000 concurrent users with ACID transactional isolation.
- Led national payment integration: high-availability Java services for the Shetab card network, and the IBM MQ messaging layer for the Chakavak national cheque-clearing system, both under strict regulatory SLAs.
- Designed high-concurrency SQL Server persistence layers (NHibernate, advanced indexing) with financial-grade consistency, and introduced Jira for structured work tracking.
- Delivered C++ POS terminal software and ATM and card-processing modules with real-time authorization and settlement, from hardware integration to backend.

Software Developer

Postbank of Iran (PBI) | Tehran, Iran | 2007 - 2009

- Developed banking and business applications in VB.NET, C#, and SQL Server, including schema design, query optimization, and JavaScript, HTML, and CSS web interfaces.

Software Developer

RASA-Rayaneh-Sabz Co. | Tehran, Iran | 2006 - 2007

- Developed client and server enterprise applications (VB.NET, SQL Server, JavaScript, HTML) through design, development, testing, and deployment.

FREELANCE CONSULTING

Software Architecture Consultant (alongside full-time role)

Amitis Hamta Co. | Tehran, Iran | 2019 - 2022

- Advised on full-stack architecture and built a secure payment panel integration for their fintech platform.

Core Banking Consultant (alongside full-time role)

Several financial institutions | Tehran, Iran | 2010 - 2017

- Architecture analysis and system design for Tejarat Bank, Kosar Financial Institute, Arman Financial Institute, and Javanan-Khayer Financial Institute, focused on regulatory compliance and high-concurrency design.

EDUCATION

B.Sc. Software Engineering

Azad University, Parand Branch (part-time while employed) | Tehran, Iran | 2008 - 2013

CERTIFICATIONS

- IBM RAG and Agentic AI Professional Certificate, Coursera (10 courses: RAG, vector databases, LangChain, LangGraph, CrewAI, AutoGen, MCP, capstone project)
- Architecting Solutions on AWS, AWS Training & Certification via Coursera (course)
- Cloud & DevOps Engineering (AWS, Docker, Kubernetes, CI/CD, Jenkins, CloudFormation)
- Cybersecurity Essentials (LFC108), Linux Foundation
- Open Source Licensing: Compliance Management and Basics for Developers, Linux Foundation

LANGUAGES

English: C1 (professional) | German: B1, working towards B2 | Persian: native